Probability - Mistakes

Test 01 (ACE)

Test-01 on Probability and Statistics-1
(Topics: Probability and Statistics: Counting (permutation and combinations), probability axioms, Sample space, events, independent events, mutually exclusive events, marginal, conditional and joint probability, Bayes Theorem.)

2. There are four cities, namely, C_1, C_2, C_3 and C_4. The cities are directly connected by four roads as shown in the picture given below, that are, C_1 is connected with C_2, C_2 is connected with C_3, C_3 is connected with C_4, and C_4 is connected with C_1. The probability of any road getting independently blocked is $\frac{1}{3}$. Let E_1 be the event of traveling from C_1 to C_3 via C_2 and E_2 be the event of traveling from C_1 to C_3 via C_4. Then, which of the following statements is correct?



A) $P(E_1 \cup E_2) = \frac{56}{81}$

B) $P(E_1 \cup E_2) = \frac{8}{9}$

C) $P(E_1/E_2) \neq P(E_1) = \frac{4}{9}$

D) $P(E_1 \cap E_2) = 0$

Correct Answer: **A** *Your Answer: C* Marks : 2 / 0.66 (-Ve)

Clearly E_1 and E_2 are independent events as they are traveling over different paths.

$\Rightarrow P(E_1 \cap E_2) = P(E_1).P(E_2)$

$P(E_1)$ = P (Not blocked from C_1 to C_2). P (Not blocked from C_2 to C_3)

$$= \frac{2}{3}.\frac{2}{3} = \frac{4}{9}$$

Similarly.

$$P(E_2) = \frac{4}{9}$$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$= \frac{4}{9} + \frac{4}{9} - \frac{4}{9}.\frac{4}{9} = \frac{56}{81}$$

$P(E_1 \cap E_2) = P(E_1) . P(E_2)$ ($\because E_1$ and E_2 are independent)

$$= \frac{4}{9}.\frac{4}{9} = \frac{16}{81}$$

$P(E_1/E_2) = P(E_1) = \frac{4}{9}$ ($\because E_1$ and E_2 are independent)

9. In a sample of 100 heart patients, each patient has 80% chance of having a heart attack without medicine X. It is clinically known that medicine X reduces the probability of having a heart attack by 50%. Medicine X is taken by 50 of these 100 patients. The probability that a randomly selected patient, out of the 100 patients, takes medicine X and has a heart attack is

A) 40%

B) 60%

C) 20%

D) 30%

Correct Answer: **C** *Not Attempted :* Marks : 2 / 0.66 (-Ve)

Let H be the event that the patient having Heart attack and X be the event that the Patient takes medicine X.

Given $P(H/\overline{X}) = 0.8$ and P (H/X) = 50% of 80%

$= 0.4$

$P(H \cap X) = P(H/X)\,P(x)$

$= (0.4)\,(0.5)$

$= 0.2$ or 20%

11. In a given day in the rainy season, it may rain 70% of the time. If it rains, chance that a village fair will make a loss on that day is 80%. However, if it does not rain, chance that the fair will make a loss on that day is only 10%. If the fair has not made a loss on a given day in the rainy season, what is the probability that it has not rained on that day?

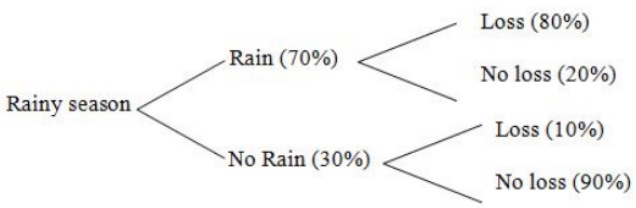
A) 3/10

B) 9/11

C) 14/17

D) 27/41

Correct Answer: **D** *Not Attempted :* Marks : 2 / 0.66 (-Ve)



$$P(\text{Not rained/No loss}) = \frac{(0.3)(0.9)}{(0.7)(0.2)+(0.3)(0.9)} = \frac{27}{41}$$

14. A diagnostic test for a certain disease is 90% accurate. That is, the probability of a person having (respectively, not having) the disease tested positive (respectively, negative) is 0.9. Fifty percent of the population has the disease. What is the probability that a randomly chosen person has the disease given that the person tested negative?

Correct Answer: **0.1** Answer Range: **(0.1 , 0.1)** *Not Attempted :*

Person tested negative
- Person not having disease and test accurately resulted negative
- Person having disease and test inaccurately resulted negative

$$P(\text{Person has the disease/Tested Negative}) = \frac{0.5(0.1)}{0.5(0.1)+0.5(0.9)} = 0.1$$

15. There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has atleast one Indian?

A) 56

B) 52

C) 48

D) 44

Correct Answer: **A** *Your Answer: B* Marks : 2 / 0.66 (-Ve)

The number of subgroups where there is atleast one Indian is given by

$$\sum_{n=0}^{3}\left(\sum_{k=1}^{3} {}^{3}C_{k}\right){}^{3}C_{n}$$

$$=\left({}^{3}C_{1}+{}^{3}C_{2}+{}^{3}C_{3}\right)\left({}^{3}C_{0}+{}^{3}C_{1}+{}^{3}C_{2}+{}^{3}C_{3}\right)$$

$$=(3+3+1)(8)=56.$$

Note:

Since, atleast one Indian should be present k is varied from 1 to 3.

Since, any number of Chinese can be present n is varied from 0 to 3.

Test 02 (ACE)

Test-02 on Probability and Statistics-2 (Topics: conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, Continuous random variables).

If the inter arrival time is exponential and 8 customers per hour arrive in a bank, then the probability of no arrival of customer during a period of 30 minutes is_____.
[Round off to three decimal places]

Correct Answer: **0.018** Answer Range: **(0.017 , 0.019)** *Not Attempted :* Marks : 2 / 0 (-Ve)

If inter arrival time is the Exponential men number of arrivals follows Poisson distribution.

8 customer's _____1 hour

4 Customers_____30 minutes

$\Rightarrow \lambda = 4$

$$p(\text{no customer}) = \frac{e^{\lambda}\lambda^0}{0!} = e^{-4} = 0.018$$

Let X_1, X_2, X_3, X_4 be independent exponential random variables with mean 1, 1/2, 1/3, 1/4, respectively. Then Y = min (X_1, X_2, X_3, X_4) has exponential distribution with mean equal to _____.

Correct Answer: **0.1** Answer Range: **(0.1 , 0.1)** *Not Attempted :* Marks : 2 / 0 (-Ve)

If X_1, X_2, X_3 and X_4 are independent exponential Random variable with mean $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}$ and $\frac{1}{\lambda_4}$ then Y = min (X_1, X_2, X_3, X_4) is also exponential with mean $\frac{1}{\lambda_1+\lambda_2+\lambda_3+\lambda_4}$

$$= \frac{1}{1+2+3+4} = \frac{1}{10} = 0.1$$

Let X_1 and X_2 be independent geometric random variables with the same probability mass function given by $P(X = k) = p(1-p)^{k-1}$, k = 1, 2,…. Then the value of $P(X_1 = 2|X_1 + X_2 = 4)$ correct up to three decimal places is _____.

Correct Answer: **0.333** Answer Range: **(0.332 , 0.334)** *Not Attempted :* Marks : 2 / 0 (-Ve)

$X_1 + X_2 = 4$

$1 + 3 = 4$ $\quad P(X_1 = 1).\ P(X_2 = 3) = (P(1-P)^0)\ (P(1-P)^2)$

$2 + 2 = 4$ $\quad P(X_1 = 2).\ P(X_2 = 2) = (P(1-P)^1)\ (P(1-P)^1)$

$3 + 1 = 4$ $\quad P(X_1 = 3).\ P(X_2 = 1) = (P(1-P)^2)\ (P(1-P)^0)$

$$= \frac{P^2(1-P)^2}{P^2(1-P)^2 + P^2(1-P)^2 + P^2(1-P)^2} = \frac{1}{3}$$

Test 01 (RBR)

Test-02 on Probability and Statistics-2 (Topics: conditional expectation and variance, mean, median, mode and standard deviation, correlation, and covariance, random variables, discrete random variables and probability mass functions, uniform, Bernoulli, binomial distribution, Continuous random variables).

Given two independent events A and B such that P(A) =0.3, P(B) = 0.6 and P($A' \cap B'$) is

- 0.9
- 0.18
- 0.28
- None of the above

Full Solution

Given, P(A) = 0.3 and P(B) = 0.6

P(A') = 1 - P(A) = 1 - 0.3

P(A') = 0.7

P(B') = 1 - P(B) = 1 - 0.6

P(B') = 0.4

P(A' ∩ B') = P(A') · P(B')

= 0.7 × 0.4

= 0.28

The value of P(A' ∩ B') is 0.28

If A and B are two mutually exclusive events, then which of the following are possible: [MSQ]

- $P(A) \leq P(\bar{B})$
- $P(B) \leq P(\bar{A})$
- $P(A) < P(B)$
- $P(A) > P(B)$

A and B are two mutually exclusive events, $P(A \cap B) = 0$

$P(A \cup B) = P(A) + P(B)$

$P(A \cup B) \leq 1$

$P(A) + P(B) \leq 1$

$P(A) = 1 - P(B)$ or $P(B) = 1 - P(A)$

$P(A) \leq P(\overline{B})$ or $P(B) \leq P(\overline{A})$

Assertion :For any two events A and B, $P(\bar{A} \cap B) = P(B) - P(A \cap B)$. Reason: $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive events.

[MSQ]

- Both Assertion & Reason are individually true & Reason is correct explanation of Assertion
- Both Assertion & Reason are individually true but Reason is not the ,correct (proper) explanation of Assertion
- Assertion is true but Reason is false
- Assertion is false but Reason is true

Test 02 (RBR)

Conditioonal expectation and variance, mean, median, mode and standard deviation, correlation, and covariance.

Time Taken: 112 secs

The correlation coefficient between two variables X and Y is 0.4. The correlation coefficient between 2X and (-Y) will be

0.4

-0.8

-0.4

0.8

Full Solution

The correlation coefficient between two variables X and Y = 0.4

The correlation coefficient (r) is independent of origin and scale and depend on the sign of variables

The correlation coefficient between the two variables is the measure of the slope between the

variables in the regression graph. It is given that the correlation coefficient between X and Y is 0.4 and the correlation coefficient is independent of change of origin and scale but it depends on variables

The correlation coefficient between 2X and (-Y) is - 0.4

Consider two exponentially distributed random variables X and Y, both having a mean of 0.50. Let Z = X + Y and r be the correlation coefficient between X and Y. If the variance of Z equals 0, then the value of r is

1

-1

5

4

Let λ be the parameter of exponential distribution

So, for exponentially distributed random variables X & Y

Expected Value or Mean $E(X) = \frac{1}{\lambda}$

Variance: $Var\ (X) = \frac{1}{\lambda^2}$

Var (X + Y) = Var X + Var Y + 2 Cov (X, Y)

Where Cov (X, Y) = Covariance of X and Y

Correction coefficient is given by:

$$r = \frac{Cov\ (X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Given: E(X) = E(Y) = 0.5

$\frac{1}{\lambda_1} = \frac{1}{\lambda_2} = 0.5$

$\lambda_1 = \lambda_2 = 2$

$Var\ (X) = \frac{1}{\lambda_1^2} = \frac{1}{2^2} = 0.25$

$Var\ (X) = \frac{1}{\lambda_1^2} = \frac{1}{2^2} = 0.25$

$Var\ (Y) = \frac{1}{\lambda_2^2} = \frac{1}{2^2} = 0.25$

Also it is given, Z = X + Y

And Var (Z) = 0

∴ Var (X + Y) = 0

Var X + Var Y + 2 Cov (X, Y) = 0

On putting values:

0.25 + 0.25 + 2 Cov (X, Y) = 0

$Cov\ (X, Y) = -\frac{0.5}{2} = -0.25$

Correction coefficient is given by:

$$r = \frac{Cov\ (X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{-0.25}{\sqrt{0.25 \times 0.25}} = -1$$

Consider the following Random Walk of an element. Suppose $y(0)=0$; that is, $y=0$ at time $t=0$. The position of $y(n)$ depends probabilistically on its time at time n-1; $y(n)=y(n-1)+1$ with probability ½, and $y(n)=y(n-1)-1$ with probability 1/2.
Which of the following is/are true?

Probability that $y(19)=0$, is non-zero

Probability that $y(n)=0$, is 0 when n is odd.

Probability that $y(10)=0$, is $({}^{10}C_5)/2^{10}$

Probability that $y(2)=0$, is 0.5.

Statement A: Probability that $y(19)=0$, is non-zero.
For $y(19)$ to be 0, the number of steps up must equal the number of steps down. Since 19 is odd, it is impossible to have an equal number of steps up and down. Therefore, the probability that $y(19)=0$ is zero.
Statement A is false.

Statement B: Probability that $y(n)=0$, is 0 when n is odd.
As explained in Statement A, for $y(n)$ to be 0, the number of steps up must equal the number of steps down. This is only possible if n is even. Therefore, if n is odd, the probability that $y(n)=0$ is indeed zero.
Statement B is true.

Statement C: Probability that $y(10)=0$, is (10C5)/2^10.
For $y(10)$ to be 0, there must be an equal number of steps up and down, i.e., 5 steps up and 5 steps down. The number of ways to choose 5 steps up out of 10 is given by the binomial coefficient $\binom{10}{5}$. The total number of possible sequences of 10 steps is 2^{10}. Therefore, the probability is $\frac{\binom{10}{5}}{2^{10}}$.
Statement C is true.

Statement D: Probability that $y(2)=0$, is 0.5.
For $y(2)$ to be 0, there must be 1 step up and 1 step down. There are 2 possible sequences: "up, down" and "down, up". The total number of possible sequences of 2 steps is $2^2 = 4$. Therefore, the probability is $\frac{2}{4} = 0.5$.
Statement D is true.

Suppose X and Y are continuous random variables with the same variance, $\sigma 2$. Define $A=X-Y$ and $B=X+Y$. What is $Corr(A,B)$?
Choose the correct answer:

0

12

1

-12

$$
\begin{aligned}
Corr(X-Y, X+Y) &= \frac{Cov(X-Y, X+Y))}{\sqrt{Var(X-Y)Var(X+Y)}} \\
&= \frac{Cov(X,X) + Cov(X,Y) - Cov(Y,X) - Cov(Y,Y)}{\sqrt{Var(X-Y)Var(X+Y)}} \\
&= \frac{Var(X) - Var(Y)}{\sqrt{Var(X-Y)Var(X+Y)}} = 0
\end{aligned}
$$